Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-PR01

Practice exam paper for first teaching September 2019

Time: 25 minutes | Paper Reference **PRACL2/01**

Mathematics

Level 2
Section A (Non-Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
16 /16

My signature confirms that I will not discuss the content of the test with anyone.

Signature: Master Functional Skills (MFS)

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and your answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used.**
- Take the value of π to be 3.14

Information

- The total mark for this section is 16
- The marks for each question are shown in brackets
  – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1 Bill is a builder.
On Monday he made mortar mix.
He used 24 kg of sand and 5 kg of cement.

On Tuesday Bill will make the same type of mortar mix.
He will use 36 kg of sand.

How much cement does he need to make the same type of mortar mix?

(3)

Step 1: Write down the ratio of Sand to Cement

Sand : Cement = 24 kg : 5 kg

Step 2: Find the scale factor multiplier for 36 kg of sand

Scale Factor = 36 ÷ 24 = 1.5 [M1]

Step 3: Multiply cement by scale factor

Cement needed = 5 × 1.5 = 7.5 kg [M1]

Alternative unit-ratio check:

- 24 ÷ 5 = 4.8 kg sand per 1 kg cement
- 36 ÷ 4.8 = 7.5 kg cement ☐

7.5 [A1] kg

(Total for Question 1 is 3 marks)

2 (a) Write 2.71828 correct to 3 decimal places.

(1)

Examine the 4th decimal digit (ten-thousandths):

2.718 | 28 --> next digit is 2 (< 5), so round down

2.718 [A1]

Here is a formula.

$P = 3T^2$

(b) Work out the value of P when $T = 10$

(2)

Step 1: Apply BIDMAS – calculate indices (powers) first:

$T^2 = 10^2 = 10 \times 10 = 100$ [M1]

Step 2: Multiply by 3:

$P = 3 \times 100 = 300$

300 [A1]

(Total for Question 2 is 3 marks)

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3 Lizzie buys 3 clocks for a total cost of £50 at a car boot sale.

She sells 2 of the clocks for £22 each and the other clock for £20

Lizzie thinks she has made a profit of over 30% of the cost of the clocks.

Is Lizzie correct?
Show why you think this.

(4)

Step 1: Calculate total selling price of the 3 clocks

Total Income = (2 × £22) + £20 = £44 + £20 = £64 [M1]

Step 2: Calculate actual profit made

Actual Profit = Selling Price − Cost = £64 − £50 [M1] £14

Step 3: Calculate target 30% profit on original cost (£50)

30% of £50 = 0.30 × £50 = £15 [M1]

Or percentage profit = (14 ÷ 50) × 100 = 28%

Step 4: Compare actual profit with Lizzie's 30% claim

£14 < £15 (28% profit < 30% target)

Therefore, Lizzie is NOT correct.

No, profit is £14 (28%), which is less than 30% (£15) [A1]

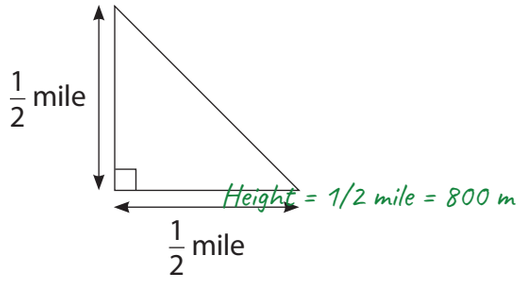
(Total for Question 3 is 4 marks)

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4 Usha is a local councillor.
She wants to write about a new housing development.

The diagram shows the space for the new development.

Base = 1/2 mile = 800 m



Usha thinks that the area of the development will be greater than the total area of 50 football pitches.

Usha knows

- a football pitch is rectangular 100 m by 50 m
- 1 mile = 1600 m.

(a) Will the area of the development be greater than the total area of 50 football pitches?

(5)

Step 1: Convert triangle dimensions from miles to metres

1 mile = 1600 m --> 1/2 mile = 1600 ÷ 2 = 800 m [MI]

Step 2: Calculate area of the triangular development

Area = (base x height) ÷ 2 = (800 x 800) ÷ 2

Area = 640,000 ÷ 2 = 320,000 m² [MI]

Step 3: Calculate area of 1 football pitch

Pitch Area = 100 m x 50 m = 5,000 m² [MI]

Step 4: Calculate total area of 50 football pitches

Total Pitch Area = 50 x 5,000 m² = 250,000 m² [MI]

(Or: 320,000 ÷ 5,000 = 64 pitches > 50 pitches)

Step 5: Compare areas and conclude

320,000 m² > 250,000 m² --> Development is larger!

Yes, the development (320,000 m²) is greater than 50 pitches (250,000 m²) [A1]

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(b) Use reverse calculations to show a check of your answer.

(1)

Reverse calculation check on football pitch area:

Check: 250,000 m² ÷ 50 = 5,000 m² ☐ **[A1]**

And: 5,000 m² ÷ 50 m = 100 m ☐

Or check triangle: 320,000 × 2 = 640,000; ☒ *640,000 = 800 m* ☒

(Total for Question 4 is 6 marks)

TOTAL FOR SECTION A = 16 MARKS

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Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-PR01

Practice exam paper for first teaching September 2019

Time: 1 hour 30 minutes | Paper Reference **PRACL2/01**

Mathematics

Level 2
Section B (Calculator)



You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
48 /48

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Signature: Master Functional Skills (MFS)

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- If your calculator does not have a π button take the value of π to be 3.14
- **Calculators may be used.**

Information

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- The total mark for this paper is 64
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Advice

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- Check your answers if you have time at the end.

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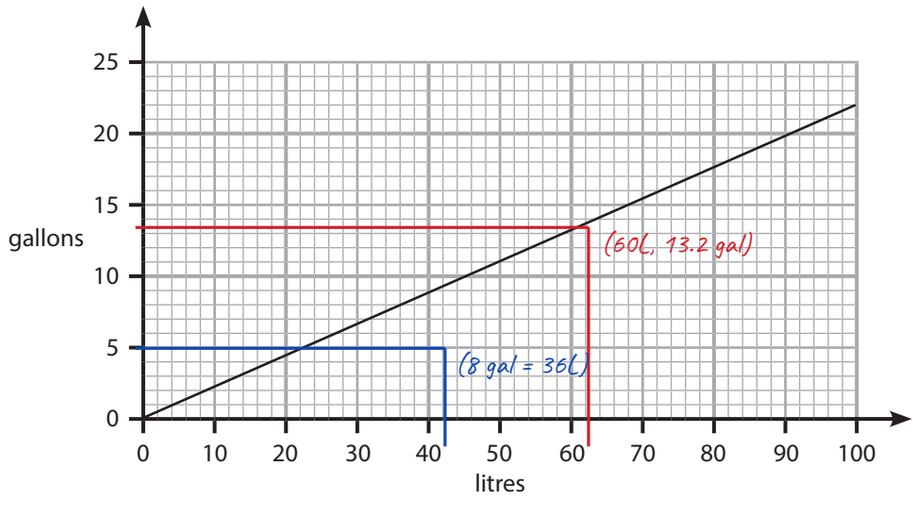
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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 This graph can be used to convert between gallons and litres.



(a) Convert 60 litres to gallons.

(1)

From graph, at 60 litres, reading = 13.2 gallons (accept 13 - 14):

13.2 [A1] gallons

One day

- Anaya used 44 litres of fuel
- Meera used 8 gallons of fuel.

Anaya used more fuel than Meera.

(b) Use the graph to work out how much more.
Remember to give units with your answer.

(2)

Step 1: Convert 8 gallons to litres using graph: 8 gallons = 36 litres

Step 2: Find difference: 44 litres – 36 litres = 8 litres [M1]

8 litres [A1]

(Total for Question 1 is 3 marks)

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2 David reads this advert on his county council website.

70% of the area of woodland in the county is native woodland.

This means there are 350 km^2 of native woodland in the county.

Step 1: Understand percentage relationship

70% of total woodland = ~~350 km²~~

Non-native woodland = 100% − 70% = 30%

Work out the area of woodland in the county that is **not** native woodland.

(3)

Step 2: Calculate 1% (unitary method) or total area

1% of woodland = 350 ÷ 70 = 5 km² [M1]

Step 3: Calculate area of non-native woodland (30%)

Non-native Area = 30 × 5 km² = 150 km² [M1]

Alternative check via total area:

- Total woodland = 100 × 5 km² = 500 km²
- Non-native = 500 − 350 = 150 km² ☐

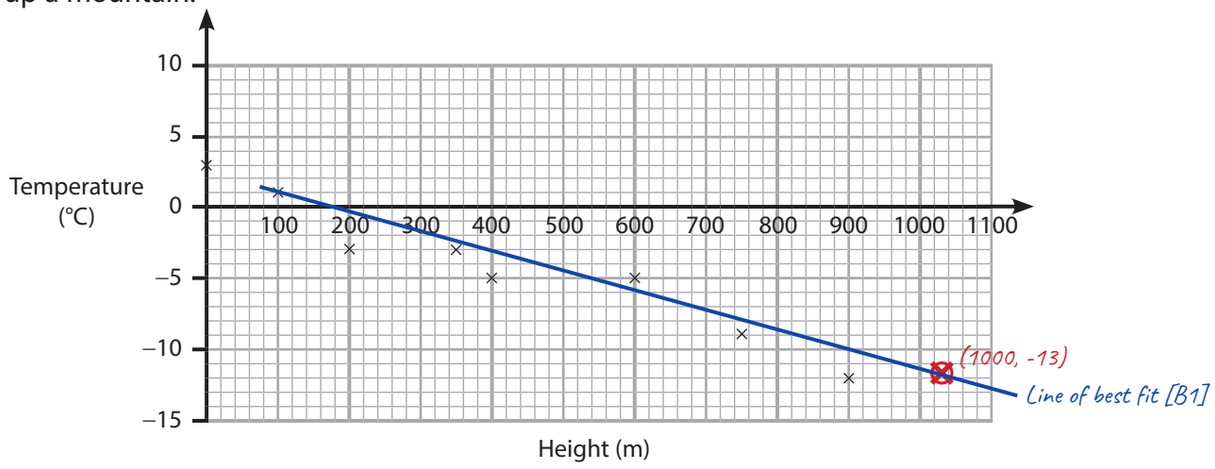
km^2

150 [A1]

(Total for Question 2 is 3 marks)

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3 The scatter diagram gives information about the temperatures at 8 different heights up a mountain.



At a height of 1000 m the temperature is −13°C.

(a) Plot this information on the scatter diagram.

(1)

Point plotted at (1000 m, −13 °C) with red cross on graph above **[A1 plot]**

(b) Draw a line of best fit on the scatter diagram.

(1)

Straight line of best fit drawn passing through the cluster of points

(c) Use the line of best fit to estimate the difference between the temperature at a height of 550 m and at a height of 950 m.

(2)

Step 1: Read temperatures from the line of best fit

- **At 550 m: Temperature ≈ −7 °C** *[M1 readings]*
- **At 950 m: Temperature ≈ −13 °C**

Step 2: Calculate difference in temperature

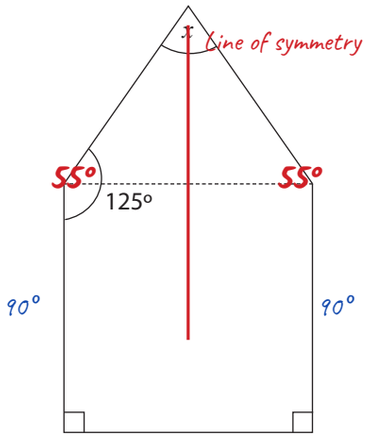
Difference = (−7) − (−13) = −7 + 13 = 6 °C **[M1]**

6 **[A1]** °C

(Total for Question 3 is 4 marks)

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4 Here is a pentagon.



The pentagon has one line of symmetry.

Work out the size of the angle marked x.

(3)

Method: Sum of interior angles of a pentagon

Sum of angles = (5 − 2) × 180° = 3 × 180° = 540° [M1]

By symmetry, upper angles = 125° each, base angles = 90° each:

x = 540° − (90° + 90° + 125° + 125°) = 540° − 430° [M1]

x = 110°

(Symmetry check: x/2 = 360° − 90° − 90° − 125° = 55° --> x = 110° ✓)

110 [A1]

°

(Total for Question 4 is 3 marks)

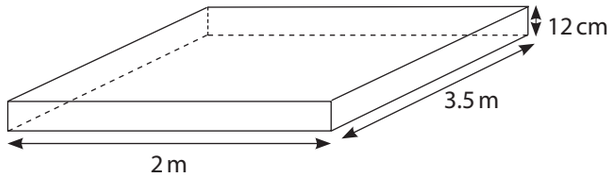
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5 Nicola wants to put a flat roof on a bike store.

The roof will be

- made of concrete
- in the shape of a cuboid as shown.

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$



Nicola wants to put a metal strip along 2 of the longest edges of the roof.

She knows

- the density of concrete is 2300 kg per m^3
- the mass of 1 metre of metal strip is 5 kg.

Work out the total mass of the concrete and the strips she wants.

(5)

Step 1: Convert thickness from cm to metres

Thickness = 12 cm ÷ 100 = 0.12 m

Step 2: Calculate volume of concrete roof (cuboid)

Volume = length × width × thickness

Volume = 3.5 m × 2 m × 0.12 m = 0.84 m^3 [M1]

Step 3: Calculate mass of the concrete

Mass = Volume × Density = 0.84 m^3 × 2300 kg/m^3

Mass of concrete = 1,932 kg [M1]

Step 4: Calculate mass of the 2 metal strips

Total metal strip length = 2 × 3.5 m = 7 m

Mass of strips = 7 m × 5 kg/m = 35 kg [M1]

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Step 5: Calculate total mass (concrete + metal strips)

Total Mass = 1,932 kg + 35 kg = 1,967 kg [M1]

Pedagogical reminder:

- Always convert all measurements to standard metres before calculating volume.
- Density formula: Mass = Density × Volume.

1967 [A1] kg

(Total for Question 5 is 5 marks)

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6 Mai has this information about 100 flowering plants in her shop.

		Stem length	
		Short	**Long**
Size of flower	**Small**	10	18
	Large	43	29

She will take a plant at random from these plants.

(a) Work out the probability that this plant will have a large flower and a long stem.

(2)

From table: Plants with Short stem and Small flower = 29 (out of 100)

Probability = 29 / 100 (= 0.29 or 29%) [M1]

29/100 [A1]

Mai will take at random a plant from the 72 plants that have a large flower.

(b) Work out the probability that this plant will have a short stem.

(1)

Condition: Restrict total to the 72 plants with a Large flower

Number of large flower plants with a short stem = 43

Probability = 43 / 72 (≈ 0.597)

43/72 [A1]

(Total for Question 6 is 3 marks)

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7 Sal works in a dress shop.
She wants to know how well the labels on the dress hangers agree with the true size of the dresses.

The table shows information about some hangers and dresses.

		True size of dress				
		10	**12**	**14**	**16**	**18**
Label on hanger	**10**	8	2	1	1	–
	12	–	9	3	1	2
	14	2	1	12	–	–
	16	1	–	1	13	2
	18	1	1	2	1	13
	Totals	12	13	19	16	17

Sal thinks that 2 in every 7 dresses are on hangers with the wrong label.

Is Sal correct?
Show clearly why you think this.

(4)

Step 1: Calculate total number of dresses in the sample

Total Dresses = 12 + 13 + 19 + 16 + 17 = 77 [M1]

Step 2: Calculate number of dresses on CORRECT hangers (diagonal)

Correct Hangers = 8 + 9 + 12 + 13 + 13 = 55 [M1]

Step 3: Calculate number of dresses on WRONG hangers

Wrong Hangers = Total − Correct = 77 − 55 = 22 dresses

Step 4: Express as a simplified fraction and compare to 2/7

Fraction with wrong label = 22 / 77

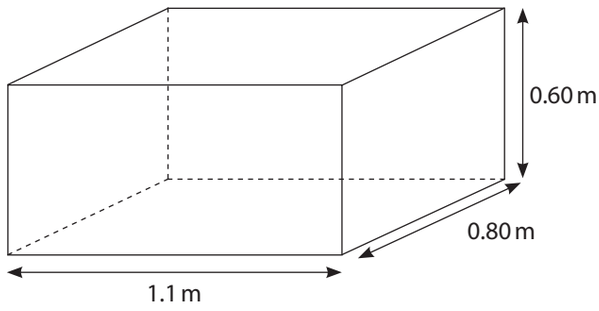
Divide by 11: (22 ÷ 11) / (77 ÷ 11) = 2/7 --> Sal is correct! [M1]

Yes, Sal is correct (22 out of 77 = 2/7) [A1]

(Total for Question 7 is 4 marks)

8 James has a contract to paint 30 identical water tanks.
He has to paint the outside surfaces of each tank, but not the top.

Each surface is rectangular.



James knows that 1 tin of paint

- is enough to cover 12 m^2 of surface
- costs £26.99

Work out the total cost of the tins of paint he will need for all 30 water tanks.

(6)

Step 1: Calculate surface area of 1 tank (base + 4 sides; no top)

- Base: 1.1 m × 0.8 m = 0.88 m^2
- 2 Front/Back faces: 2 × (1.1 m × 0.6 m) = 1.32 m^2 *[M1 area of faces]*
- 2 Left/Right faces: 2 × (0.8 m × 0.6 m) = 0.96 m^2

Surface Area of 1 Tank = 0.88 + 1.32 + 0.96 = 3.16 m^2 [M1]

Step 2: Calculate total area to paint for all 30 tanks

Total Area = 30 × 3.16 m^2 = 94.8 m^2 [M1]

Step 3: Calculate number of tins of paint needed

Tins needed = 94.8 m^2 ÷ 12 m^2/tin = 7.9 tins [M1]

You cannot buy a partial tin, so must round up to 8 whole tins!

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Step 4: Calculate total cost of 8 tins of paint at £26.99 each

Total Cost = 8 × £26.99 [M1]

Total Cost = £215.92

Summary breakdown for examiner check:

- 30 tanks × 3.16 m² = 94.8 m²
- 94.8 ÷ 12 = 7.9 tins --> 8 tins
- 8 × £26.99 = £215.92

£ 215.92 [A1]

(Total for Question 8 is 6 marks)

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9 Andros has an oil fired heating system.
In a 30-day period he used a full tank of oil at a constant rate per day.

At a different time of the year the amount of oil Andros uses per day is $\frac{1}{3}$ of the rate used in the 30-day period.

(a) How many days should a full tank of oil last at this new rate?

(2)

Step 1: Understand inverse proportion relationship

If the daily consumption rate is multiplied by 1/3,

the time the tank lasts is multiplied by the reciprocal (3).

Step 2: Calculate new number of days

Days = 30 days ÷ (1/3) = 30 × 3 = 90 days [M1]

90 [A1]

days

(b) Use reverse calculation to show a check of your answer.

(1)

Reverse calculation check:

Check: 90 days × (1/3) = 30 days ☐ [A1]

Or: 90 ÷ 3 = 30 days ☐

(Total for Question 9 is 3 marks)

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10 Lena recorded the number of late trains at a station in a day over a period of time.

She shows this information in a diagram.

Number of days										
	5									
	4		*							
	3		*	*						
	2	*	*	*	*	*				
	1	*	*	*	*	*	*			*
		1	2	3	4	5	6	7	8	9
		Number of late trains in a day								

Key: * means there were 2 days when 4 trains were late each day.

(a) For this information work out the median number of late trains in a day.

(2)

Step 1: Write all 15 recorded values in ascending numerical order

1, 1, 2, 2, 2, 2, 3, 3, 3, 4, 4, 5, 5, 6, 9 *[M1 ordered list]*

Step 2: Find the middle value position

Position = (15 + 1) ÷ 2 = 8th value

The 8th value in the list is 3.

3 **[A1]**

Lena says 'The median number of late trains in a day, from this information, is a good estimate of the average number of trains late over the period of time.'

(b) Is Lena correct?
Explain why you think this.

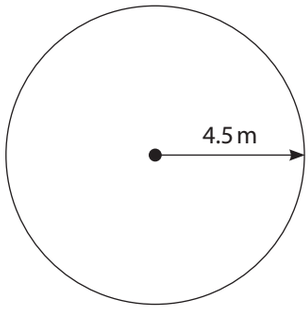
(1)

Yes, because the mean is 48 ÷ 15 = 3.2, which is very close to the median (3). **[A1]**

(Mark scheme also accepts "No, because the mode is 2".)

(Total for Question 10 is 3 marks)

11 Joanna is a landscape gardener.
She has to fill a circular space with flowers.



The radius of the circular space is 4.5 metres.

Joanna will plant 40 flowers per square metre of space.

She will plant 4 times as many red flowers as white flowers.

How many red flowers will she plant?

(5)

Step 1: Calculate area of the circular space

Area = $\pi \times r^2 = \pi \times 4.5^2$ [M1]

Area = $\pi \times 20.25 = 63.617...\ m^2$

Step 2: Calculate total number of flowers

Total flowers = $63.617...\ m^2 \times 40$ flowers/m^2 [M1]

Total flowers = $2{,}544.69... \approx 2{,}545$ flowers

Step 3: Use ratio Red : White = 4 : 1 (5 total parts)

Fraction that is red = 4 / (4 + 1) = 4/5

Red flowers = $2{,}544.69... \times (4/5) = 2{,}035.75... \approx 2{,}036$ flowers [M1] [A1]

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Step 4: Precision check across different values of π:

- Using full calculator π: $63.6173 \times 40 \times 4/5 = 2035.75$ --> 2036 red flowers
- Using $\pi = 3.14$: $63.585 \times 40 \times 4/5 = 2034.72$ --> 2035 red flowers
- Mark scheme fully accepts 2034, 2035, or 2036.

2036 [A1]

(Total for Question 11 is 5 marks)

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12 Jim owns a small business.

The table shows information about the weekly wage of the 40 workers.

Weekly wage (£)	Number of workers
320	10
370	13
420	8
470	7
520	2

Jim wants to increase the mean wage by 4%, plus £10

Jim thinks the new mean weekly wage of these workers will be more than £415

Is Jim correct?
You **must** show your working.

(6)

Step 1: Find midpoints (x) and fx products for each wage band

- £300 to <£340: midpoint £320 × 10 = £3,200 *[M1 fx table]*
- £340 to <£400: midpoint £370 × 13 = £4,810
- £400 to <£440: midpoint £420 × 8 = £3,360
- £440 to <£500: midpoint £470 × 7 = £3,290
- £500 to <£540: midpoint £520 × 2 = £1,040

Step 2: Calculate total wage bill before increase

Total Wages = 3200 + 4810 + 3360 + 3290 + 1040 [M1] £15,700

Step 3: Calculate current mean weekly wage

Current Mean = £15,700 ÷ 40 workers = £392.50 [M1]

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Step 4: Apply 4% wage rise to current mean

4% of £392.50 = 0.04 × £392.50 = £15.70 [M1]

Mean after 4% = £392.50 + £15.70 = £408.20

Step 5: Add additional £10 to mean weekly wage

New Mean = £408.20 + £10 = £418.20 [M1]

Step 6: Compare with Jim's prediction (£415)

£418.20 > £415.00 --> Jim is correct!

Yes, Jim is correct as the new mean wage is £418.20 (> £415) [A1]

(Total for Question 12 is 6 marks)

TOTAL FOR SECTION B = 48 MARKS
TOTAL FOR PAPER = 64 MARKS

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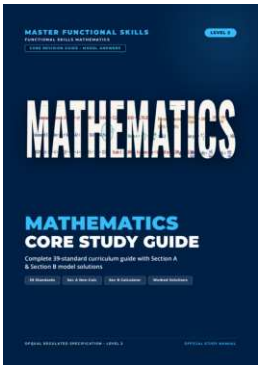
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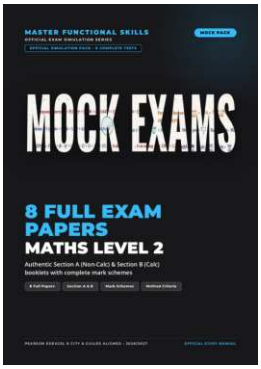


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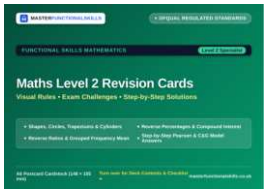


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